

July 2025 - CSE 108

Online on Threading & Networking

Section: B1/B2

Time: 45 minutes

Given an integer range $[MIN, MAX]$ (inclusive) where $MIN < MAX$, compute the sum of all prime numbers within this range using $N_THREADS$ threads running in parallel.

Each thread must:

- Work on a separate contiguous sub-range
- Compute the partial sum of primes within its assigned range
- Print its assigned range and partial sum

After all threads finish, the main thread must compute and print the final prime sum.

A helper function `is_prime(long a)` is provided below for your ease.

Thread Work Division Rule

- $TOTAL = (MAX - MIN + 1)$
- $chunk = TOTAL / N_THREADS$

Threads must be assigned ranges as follows:

- Thread t (0-based) handles:
 - $start = MIN + t * chunk$
 - $end = start + chunk - 1$ for all threads except the last
- The last thread handles:
 - $start = MIN + (N_THREADS - 1) * chunk$
 - $end = MAX$

Thus, the last thread may receive a *slightly different number* of elements.

Input Format:

`MIN MAX N_THREADS`

Constraints:

- $\text{MIN} < \text{MAX}$, $\text{N_THREADS} \geq 1$, $\text{MIN} \geq 0$

Output Format:

Each thread prints:

Thread-<thread_no>: range [<start>, <end>] partialSum = <sum>

Finally, print:

Final Sum = <final_sum>

Sample I/O:

Input 1:

1 200 5

Output1 (order may vary)

Thread-2: range [81, 120] partialSum = 802
 Thread-0: range [1, 40] partialSum = 197
 Thread-1: range [41, 80] partialSum = 594
 Thread-3: range [121, 160] partialSum = 991
 Thread-4: range [161, 200] partialSum = 1643
 Final PrimeSum = 4227

Input 2:

1 25 2

Output2 (order may vary)

Thread-0: range [1, 12] partialSum = 28
 Thread-1: range [13, 25] partialSum = 72
 Final PrimeSum = 100

Task	Marks
Correct thread creation using N_THREADS	3
Correct work division into contiguous ranges	3
Correct partial sum per thread	2
Correct final sum	2
Total	10

The method for prime check

```
import java.util.*;
static boolean is_prime(long a) {
    if (a < 2) return false;
    if (a == 2) return true;
    if (a % 2 == 0) return false;

    long r = (long) Math.sqrt(a);
    for (long i = 3; i <= r; i += 2) {
        if (a % i == 0) return false;
    }
    return true;
}
```